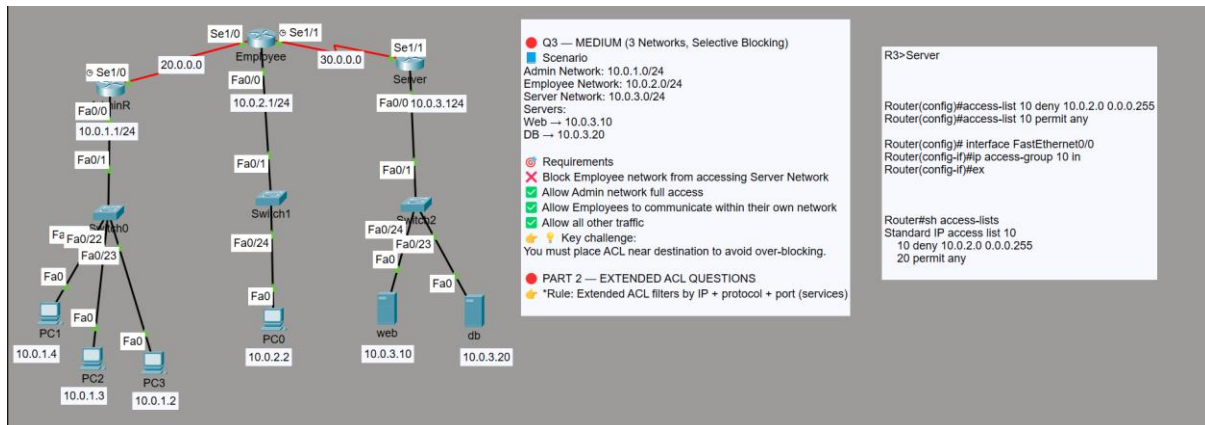


ACL

{do routing first}

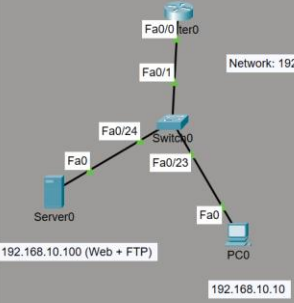
STANDARD ACL

Q1)



Extended

Q1)



Network: 192.168.10.0/24

192.168.10.100 (Web + FTP)

192.168.10.10

Q1 — EASY (Single Network, Service Control)

Scenario

Network: 192.168.10.0/24

PC1 → 192.168.10.10

Server → 192.168.10.100 (Web + FTP)

Requirements

- Block FTP (21) from PC1
- Allow HTTP (80) from PC1
- block telnet
- Allow all other traffic

```
Router(config)#access-list 100 deny tcp host 192.168.10.10 host 192.168.10.100 eq 21
Router(config)#access-list 100 deny tcp host 192.168.10.10 host 192.168.10.100 eq 23
Router(config)#access-list 100 permit tcp host 192.168.10.10 host 192.168.10.100 eq 80
Router(config)#access-list 100 permit ip any any

Router(config)#interface fa0/0
Router(config-if)#ip access-group 100 in

VERIFY

Router#show access-list

access-list 100 deny tcp host 192.168.10.10 host 192.168.10.100 eq ftp
access-list 100 deny tcp host 192.168.10.10 host 192.168.10.100 eq telnet
access-list 100 permit tcp host 192.168.10.10 host 192.168.10.100 eq www
access-list 100 permit ip any any
Router#

Router#show run

interface FastEthernet0/0
ip address 192.168.10.1 255.255.255.0
ip access-group 100 in
duplex auto
speed auto
```

Rule

Meaning

deny tcp PC1 → Server eq 21

Block FTP

permit tcp PC1 → Server eq 80

Allow HTTP

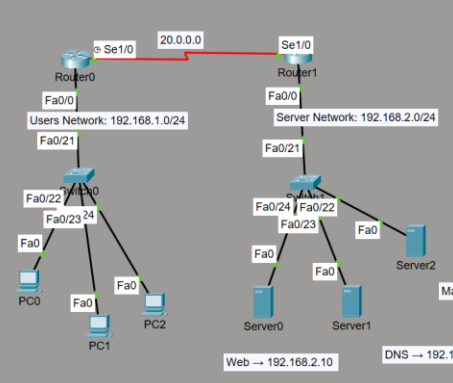
deny tcp any any eq 23

Block Telnet globally

permit ip any any

Allow everything else

Q2



20.0.0.0

Se1/0

Se1/0

Router0

Router1

Users Network: 192.168.1.0/24

Server Network: 192.168.2.0/24

Web → 192.168.2.10

DNS → 192.168.2.20

Mail → 192.168.2.30

Q2 — MEDIUM (Multi-Network + Multiple Servers)

Scenario

Users Network: 192.168.1.0/24

Server Network: 192.168.2.0/24

Servers:

- Web → 192.168.2.10
- DNS → 192.168.2.20
- Mail → 192.168.2.30

Requirements

- Block HTTP (80) to Web Server
- Allow DNS (53)
- Block SMTP (25) to Mail Server
- Allow Ping (ICMP)
- Allow all other traffic

```
Router#sh access-lists
Extended IP access list 100
10 deny tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq www
20 permit udp 192.168.1.0 0.0.0.255 host 192.168.2.20 eq domain
30 deny tcp 192.168.1.0 0.0.0.255 host 192.168.2.30 eq smtp
40 permit icmp 192.168.1.0 0.0.0.255 any
50 permit ip any any
Router#

Router(config)#access-list 100 deny tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 80
Router(config)#access-list 100 permit udp 192.168.1.0 0.0.0.255 host 192.168.2.20 eq 53
Router(config)#access-list 100 deny tcp 192.168.1.0 0.0.0.255 host 192.168.2.30 eq 25
Router(config)#access-list 100 permit icmp 192.168.1.0 0.0.0.255 any
Router(config)#access-list 100 permit ip any any

Router(config)#interface FastEthernet0/0
Router(config-if)#ip access-group 100 in
Router(config)#ex
```